

CONSISTENT QUANTIZATION OF NONLINEAR SIGMA MODELS COUPLED TO WEYL FERMIONS

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Received 29 August 1988

Revised 31 October 1988

We discuss the quantization of 2-dimensional nonlinear sigma models defined in G/H spaces using the path-integral approach. We show that even when anomalies are present, a careful definition of the quantum effective action leads to a consistent quantum theory. The correct treatment of the H degrees of freedom uncovers the presence of a Wess-Zumino action and the anomaly is absorbed.

1. Introduction

Nonlinear sigma models with boson fields coupling to massless fermions¹ are relevant in a number of contexts: low-energy approximations to chiral gauge theories,² supergravity,³ string theories.⁴

It has been observed⁵ and extensively discussed^{6–10} that nonlinear sigma models in which the scalar manifold is a coset space G/H , may show up anomalies if fermions are not in anomaly-free representations of the group H (the isotropy group). Moreover, it has been shown^{8–10} that even if anomalies exist, they can be cancelled in certain cases by a set of local counterterms. In this last case, quantization follows the usual (nonanomalous) lines although scalars may gain mass at the quantum level.¹

We explore in this paper, an alternative approach to quantization of sigma models, inspired in Polyakov¹¹ and Faddeev-Shatashvili¹² ideas on quantization when anomalies are present. This approach has already shown to lead to consistent quantum theories in the case of gauge^{13–15} and gravitational¹⁶ models coupled to Weyl fermions.

The idea behind this approach, when applied to the sigma model case is the following. In the path-integral framework one should start from a generating functional Z defined as an integral over the full symmetry group G , although in the classical Lagrangian (with scalars $\phi \in G/H$ coupled to fermions through a canonical connection $A_\mu^h[\phi]$ depending on ϕ and h , $h \in H$) the h -dependence can be trivially eliminated. When Dirac fermions are present, h -fields can be gauged away also at the quantum level and then, one can look at Z as an integral over the orbit space G/H . On the contrary, when Weyl fermions are present—and because of their ability to create gauge noninvariant interactions—this is not the case anymore and one has to reconsider the standard procedure for quantization.

The outcome is that h fields enter a nontrivial effective action, the Wess-Zumino action being a part of it; moreover, the anomaly is absorbed. The same phenomenon happens with the Liouville action in the quantization of strings¹¹ and with the Wess-Zumino action in quantization of gauge and gravitational theories coupled to Weyl fermions.^{13–16}

It is important to note that in this approach we do not have to add the Wess-Zumino term by hand since it emerges naturally in the process of quantization. One can however, make contact with the approach developed in Refs. 8–10, in which anomalies are eliminated by the addition of appropriate counterterms, as we explain at the end of this work.

Our paper is organized as follows. In Sec. 2, we discuss the model which we take for simplicity in 2 space-time dimensions (the quantization approach we present is intended to work in arbitrary number of dimensions). We describe in some details the classical symmetries and associated currents. In Sec. 3, we show how anomalies arise and compute them explicitly using Fujikawa's approach. In Sec. 4, we describe how consistent quantization can be implemented in the path-integral approach even if anomalies are present. Sec. 5 is devoted to the conclusions. An Appendix contains details of the calculations.

2. The Model

Let us consider a model with N_B scalar fields ϕ^i ($i = 1, 2, \dots, N_B$) lying on the coset space G/H , with G a Lie Group and $H \subset G$ the isotropy group; $N_B = \dim G - \dim H$. Scalar fields are coupled to Weyl fermions transforming according to some representation of H .

We call H_a ($a = 1, 2, \dots, \dim H$) the H generators and T_i the other generators of G ($i = 1, 2, \dots, N_B$). If the Lie algebra of G admits a reductive splitting, the commutation relations take the form:

$$\begin{aligned} [H_a, H_b] &= f_{ab}^c H_c \\ [H_a, T_i] &= f_{ai}^j T_j \\ [T_i, T_j] &= f_{ij}^a H_a + f_{ij}^k T_k. \end{aligned} \quad (2.1)$$

The sigma model coupled to Weyl fermions can be formulated starting from the Lagrangian⁷ (in 2-dimensional Minkowski space):

$$\mathcal{L} = \text{tr}(D_\mu g)^+ D^\mu g - i\bar{\psi} \not{D} \psi \quad (2.2)$$

Here $g \in G$, D_μ is a covariant derivative,

$$D_\mu g = \partial_\mu g - g A_\mu^a H_a \quad (2.3a)$$

$$D_\mu \psi = \partial_\mu \psi - A_\mu^a h_a \psi \quad (2.3b)$$

with A_μ an auxiliary field taking values in the Lie algebra of H (we call h_a the H generators in the fermionic representation). For definiteness we consider right-handed fermions, $\psi = \gamma_5 \psi$.

Using the equations of motion for A_μ and the following parametrization for g :²

$$g = g^* h \quad (2.4)$$

with

$$g^* = e^{i\phi}; \quad \phi = \phi^i T_i \quad (2.5)$$

$$h = e^{i\eta}; \quad \eta = \eta^a H_a \quad (2.6)$$

Lagrangian (2.2) becomes:

$$\mathcal{L} = g_{ij}[\phi] \partial_\mu \phi^i \partial^\mu \phi^j - i \bar{\psi} \not{D}[A^h[\phi]] \psi \quad (2.7)$$

where $g_{ij}[\phi]$ is a G -invariant metric on G/H :

$$g_{ij}[\phi] = \text{tr}(e^{-i\phi} \partial_i e^{i\phi}|_{G/H} e^{-i\phi} \partial_j e^{i\phi}|_{G/H}) \quad (2.8)$$

(We indicate with the subscript G/H projection on G/H).

Concerning the covariant derivative $D_\mu[A]$, it takes the form

$$D_\mu[A] = \partial_\mu + A_\mu[\phi] \quad (2.9)$$

with A_μ the canonical connection:¹

$$A_\mu^a[\phi] h_a = e^{-i\phi} \partial_\mu e^{i\phi}|_H \quad (2.10)$$

and

$$A_\mu^h = h^{-1} A_\mu h + h^{-1} \partial_\mu h. \quad (2.11)$$

Note that in the classical Lagrangian (2.7) h does not represent a true degree of freedom since it can be gauged away making the rotation:

$$\begin{aligned} \psi &\rightarrow h^{-1} \psi \\ \bar{\psi} &\rightarrow \bar{\psi} h \end{aligned} \quad (2.12)$$

leading to:

$$\mathcal{L} = g_{ij}[\phi] \partial_\mu \phi^i \partial^\mu \phi^j - \bar{\psi} \not{D}[A[\phi]] \psi. \quad (2.13)$$

However, at the quantum level, transformation (2.12) may have an associated non-trivial Fujikawa's Jacobian¹⁷ $J[h]$. It has become clear after the pioneering paper of Polyakov¹¹ on string quantization, that degrees of freedom which are decoupled at the classical level may reappear after quantization playing a physical role. This is precisely the case when $J \neq 1$. We shall then take (2.7) (and not (2.13)) as the starting point for quantization.

At the classical level Lagrangian (2.13) is invariant under the global G transformation:

$$e^{i\phi} \rightarrow K e^{i\phi} f^{-1}[\phi, K] = e^{i\phi'} \quad (2.14)$$

$$\psi \rightarrow f\psi; \quad \bar{\psi} \rightarrow \bar{\psi} f^{-1}, \quad (2.15)$$

with $K \in G$ (global transformation) and $f \in H$ a compensating transformation depending on ϕ and K , which projects $K e^{i\phi}$ back to the form (2.5).

Under transformation (2.14) A_μ changes as

$$A_\mu[\phi] \rightarrow A_\mu^{f^{-1}}[\phi]. \quad (2.16)$$

Infinitesimally, transformations (2.14)–(2.15) can be written as follows:

(i) when $K \in H$,

$$K \simeq 1 + \varepsilon^a H_a \quad (2.17)$$

$$\delta\phi^i \simeq \varepsilon^a f_{aj}^i \phi^j \quad (2.18)$$

$$\delta\psi \simeq i\varepsilon^a h_a \psi \quad (2.19)$$

i.e., it is a linear transformation.

(ii) when K is generated by T_i 's:

$$K \simeq 1 + \varepsilon^i T_i \quad (2.20)$$

$$\delta\phi^i \simeq \varepsilon^i - \frac{1}{2}\varepsilon^j f_{jk}^i \phi^k + \frac{1}{4}\varepsilon^j f_{jk}^a f_{al}^i \phi^k \phi^l \quad (2.21)$$

$$\delta\psi \simeq -\frac{i}{2}\varepsilon^i \phi^j f_{ij}^a h_a \psi \quad (2.22)$$

i.e., the G -action generates in this case a nonlinear G -realization.

It is interesting to note that when the h -field is kept in the Lagrangian, global G transformations can be viewed as ϕ -transformations accompanied by fermionic rotations:

$$e^{i\phi} \rightarrow K e^{i\phi} f^{-1} \quad (2.23)$$

$$\psi \rightarrow h^{-1} f h \psi \quad (2.24)$$

or as transformations of ϕ and h fields, not affecting fermions.

$$e^{i\phi} \rightarrow K e^{i\phi} f^{-1} \quad (2.25)$$

$$h \rightarrow h' = f h \quad (2.26)$$

One can easily find the conserved current associated with the G -invariance of the model using Noether method. The answer is:

$$j_\mu^a = 2f^{akj} g_{ik}[\phi] (\partial_\mu \phi^i) \phi^j - i f^{akj} \phi^j \bar{\psi} \gamma_\mu (e^{-i\phi} \partial_\mu e^{i\phi})_k \psi + \bar{\psi} \gamma^\mu h^a \psi. \quad (2.27)$$

3. Anomalies

As we stated above, the nonlinear sigma model coupled to Weyl fermions can be afflicted with anomalies⁵ breaking the invariance under the full isometry group G .

Anomalies affect the sigma model quantization at two stages. First, elimination of the gauge degrees of freedom represented by h , using transformation (2.12) is not possible: although h disappears from Lagrangian (2.7), it reappears through the non-trivial Fujikawa Jacobian associated to (2.12). This is precisely analogous to the appearance of a Liouville action in string quantization:¹¹ due to conformal anomalies, for $d \neq 26$, the classically decoupled degree of freedom reappears as physical after quantization.

There is a second place where anomalies can affect quantization even if one starts from the Lagrangian (2.13) (i.e., one fixes the gauge classically and then quantize the model). Indeed, since the existence of anomalies is due to the impossibility of regularizing ultraviolet divergencies without affecting the corresponding symmetry, the quantum Noether theorem for the Lagrangian (2.13) will show an anomalous term indicating that G -symmetry is lost.

In order to analyze anomalies in the sigma model, we consider the fermionic path-integral:

$$\exp(i\Gamma[\phi]) \equiv \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp\left(\int \bar{\psi} \mathcal{D}[\phi] \psi d^2x\right) = \det \mathcal{D}[A[\phi]]. \quad (3.1)$$

Under an infinitesimal rotation:

$$\begin{aligned} \psi &\rightarrow e^{i\varepsilon} \psi \simeq \psi + i\varepsilon \psi \\ \bar{\psi} &\rightarrow \bar{\psi} e^{-i\varepsilon} \simeq \bar{\psi} - i\bar{\psi} \varepsilon \end{aligned} \quad (3.2)$$

with ε taking values in the Lie algebra of H , (3.1) changes according to:

$$\det \mathcal{D}[A[\phi]] = J[\varepsilon] \det \mathcal{D}[A^{e^{i\varepsilon}}[\phi]]. \quad (3.3)$$

Both determinants in (3.3) are ill-defined quantities requiring some regularization. If this regularization affects the classical symmetry, a nontrivial Jacobian J arises. Being Weyl fermions, operator $\mathcal{D}[A]$ does not have a well-defined eigenvalue problem since it maps positive chirality spinors into negative chirality ones. Usually, this problem is overcome by adding free fermions with opposite chirality¹⁸ but then the modified Dirac operator is no more a covariant derivative and $\det \mathcal{D}$ is not forced to be H -invariant.

We sketch in an Appendix how $J[\varepsilon]$ can be calculated (using, for example the heat-kernel regularization technique). The answer is:

$$\log J[\varepsilon] = i \int d^2x \omega(\varepsilon, A[\phi]) \quad (3.4)$$

with

$$\omega(\varepsilon, A[\phi]) = \frac{1}{4\pi} \text{tr}(\partial_- A_+ - a \partial_+ A_- + a[A_-, A_+])\varepsilon(x). \quad (3.5)$$

Here, $A_{\pm} = A_0 \pm A_1$ and a is an undetermined parameter appearing due to the regularization ambiguities (see Appendix). The consistent anomaly can be obtained if one fixes $a = 0$.

The anomaly polynomial $\omega(\varepsilon, A)$ can also be determined algebraically and then related to the gauge-variation of the Chern-Simons three-form $Q_3[A]$:^{9,10}

$$\delta_\varepsilon Q_3[A] = dQ_2^1(\varepsilon, A), \quad (3.6)$$

where Q_2^1 is the anomaly polynomial in differential geometry notation,

$$\log J[\varepsilon] = i \int Q_2^1(\varepsilon, A). \quad (3.7)$$

If one considers finite fermion rotations,

$$\begin{aligned} \psi' &= f\psi \\ \bar{\psi}' &= \bar{\psi}f^{-1} \end{aligned} \quad (3.8)$$

with $f = e^{i\eta} \in H$, the corresponding Jacobian can be evaluated from the infinitesimal one:

$$\log J[f, A] = i \int_0^1 dt \int d^2x \omega(\eta_t, A^{f_t}). \quad (3.9)$$

Here f_t is a one-parameter family of H -transformations interpolating between the identity and $f(x)$ as t ranges from 0 to 1; $\eta_t = -if_t^{-1}\partial_t f_t$. In the Appendix we show

that one gets for $J[f]$:

$$\log J[f, A] = \omega[f] + \frac{1}{4\pi} \text{tr} \int d^2x [A_+ f \partial_- f^{-1} + a(A_- A_+^f - A_- A_+)] \quad (3.10)$$

with $\omega[f]$ the Wess-Zumino-Witten action:

$$\omega[f] = \frac{1}{8\pi} \text{tr} \int d^2x \partial_\mu f^{-1} \partial_\mu f + \frac{1}{12\pi} \text{tr} \int_B d^3y \varepsilon^{ijk} f_i^{-1} \partial_i f_t f_t^{-1} \partial_j f_t f_t^{-1} d_k f_t. \quad (3.11)$$

One can derive from Jacobian (3.4)–(3.5) the sigma model anomaly using the Noether method in the path-integral approach. Classically, the fermionic current J_μ^a ,

$$J_\mu^a = \bar{\psi} \gamma_\mu h_a \psi \quad (3.12)$$

is covariantly conserved. At the quantum level one can check if this conservation still holds by studying the behavior of the fermionic path-integral under the local H -transformation:

$$\begin{aligned} \psi &\rightarrow e^{ie(x)} \psi \\ \bar{\psi} &\rightarrow \bar{\psi} e^{-ie(x)}. \end{aligned} \quad (3.13)$$

Using Eq. (3.3) one gets:

$$\det \mathcal{D}[A^h] = J[\varepsilon] \int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{\{\bar{\psi} \mathcal{D}[A^h] \psi + i\varepsilon^a D_\mu^{ab} J^{\mu b}\} d^2x} \quad (3.14)$$

Since the left-hand side of (3.14) does not depend on ε , one gets, after differentiation,

$$\langle D_\mu^{ab} J_b^\mu \rangle = i \left. \frac{\delta \log J[\varepsilon]}{\delta \varepsilon^a} \right|_{\varepsilon=0} \quad (3.15)$$

or

$$\langle D_\mu^{ab} J_b^\mu \rangle = -\frac{1}{4\pi} (\partial_- A_+^a - a D_+^{ab} A_-^b). \quad (3.16)$$

Now, since $\Gamma[\phi]$ changes under (2.20)–(2.22) in the form:

$$\delta \Gamma = i \int \varepsilon^i \phi^j f_{ij}^a \langle D_\mu^{ab} J^{\mu b} \rangle d^2x \quad (3.17)$$

the result (3.16) implies an anomalous behavior of Γ for any value of a .

Concerning global rotations in the H -subgroup of G , one has, instead of (3.17):

$$\delta_{\text{global}} \Gamma = i\varepsilon^a \int \langle D_\mu^{ab} J_b^\mu \rangle d^2x. \quad (3.18)$$

We see from (3.16) that for $a = 0$

$$\delta_{\text{global}} \Gamma = 0 \quad (3.19)$$

Concerning the $a \neq 0$ case, one can find adequate counterterms (of the form $\int A_+ A_- d^2x$) which, added to the original Lagrangian, lead to (3.16). One then concludes that rigid H -rotations are not anomalous.

4. Consistent Quantization of Anomalous Models

We have seen that if fermions transform in anomalous representations of H , the path-integral (3.1) is not invariant under G -isometries. If one defines the quantum effective action as given by this path-integral, one then faces the problem of defining a consistent quantum theory.

It is known however that a careful treatment of gauge degrees of freedom naturally leads to an alternative definition of the effective action. In the case of gauge and gravitational models coupled to Weyl fermions,^{13,16} chiral bosons²⁰ and other models presenting anomalies, this definition allows a consistent quantization; we shall explain in this section how an analogous treatment can be implemented for the case of nonlinear sigma models.

Consider the complete generating functional for the model with dynamics defined by Lagrangian (2.7):

$$Z = \int \mathcal{D}\phi \mathcal{D}h \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[i \int (g_{ij} \partial_\mu \phi^i \partial^\mu \phi^j - i\bar{\psi} \not{D}[A^h] \psi) d^2x \right] \quad (4.1)$$

were Γ (given by (3.1)) nonanomalous, h would be trivially eliminated from (4.1) using for example, the Faddeev Popov (FP) procedure. Due to the presence of anomalies, this is no more possible and h remains as a dynamical field. Indeed, consider the FP resolution of the identity:

$$1 = \int \mathcal{D}f \delta(hf - 1). \quad (4.2)$$

Inserting (4.2) in (4.1) and performing the change of variables:

$$\begin{aligned} h &\rightarrow hf \\ \psi &\rightarrow f^{-1}\psi \\ \bar{\psi} &\rightarrow \bar{\psi}f \end{aligned} \quad (4.3)$$

we get:

$$Z = \int \mathcal{D}\phi \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{D}f J^{-1}[f^{-1}, A[\phi]] \exp \left[i \int d^2x \mathcal{L} \right] \quad (4.4)$$

where the Fujikawa Jacobian J is given by (3.10).

We then see that the natural effective action for ϕ -fields is, in complete analogy with gauge theories:¹³

$$\exp[i\Gamma_{\text{eff}}[\phi]] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{D}f J^{-1}[f^{-1}, A[\phi]] \exp \left(\int \bar{\psi} \not{D} \psi d^2x \right). \quad (4.5)$$

Only in the nonanomalous case ($J = 1$) Γ_{eff} coincides with Γ . Otherwise, the f -field, taking values in H remains playing a dynamical role, at an equal foot with fermions and ϕ -scalars, with a Wess-Zumino type action.

As we stated above, gauge and gravitational theories can be treated, when coupled to Weyl fermions, in an analogous form. In fact this point of view for quantization of anomalous models is inspired in Polyakov approach to string quantization¹¹ and was suggested for gauge theories by Faddeev and Shatashvili.¹²

It is easy to check that Γ_{eff} as defined in (4.7), is invariant under G -isometries. Indeed, the one-cocycle property:

$$\log J[h, A] + \log J[f, A^h] = \log J[hf, A] \quad (4.6)$$

guarantees that the change in the fermionic measure under G -rotations,

$$\mathcal{D}\bar{\psi} \mathcal{D}\psi \rightarrow J^{-1}[h^{-1}, A] \mathcal{D}\bar{\psi} \mathcal{D}\psi \quad (4.7)$$

is compensated by the change in the Wess-Zumino action coming from J :

$$\exp(\log J^{-1}[f^{-1}, A]) \rightarrow \exp(\log J^{-1}[f^{-1}, A^{h^{-1}}]) \quad (4.8)$$

(one also uses $\mathcal{D}fh = \mathcal{D}f$).

It is important to stress the difference between the approach described above and the one presented in Ref. 21 following Faddeev-Shatashvili idea. In the latter approach, a Wess-Zumino term is added by hand so as to cancel the anomaly while in the present treatment it naturally emerges in the process of quantization. As pointed out in Ref. 22, cancellation of the fermionic measure variation with that coming from the Wess-Zumino action is not automatic and depends on a consistent regularization of fermionic determinants. In the formulation we presented, both variations have the same origin and come from determinants regularized in a unique form. This is the reason why cancellation is ensured in our treatment, as can be checked by explicit calculation.

The G -invariance of the effective action (4.5) can be seen even more straightforwardly

if one integrates out fermions:

$$\exp[i\Gamma_{\text{eff}}[A[\phi]]] = \int \mathcal{D}f \det \mathcal{D}[Af]. \quad (4.9)$$

It is now obvious that $\Gamma_{\text{eff}}[A] = \Gamma_{\text{eff}}[A^h]$. As explained in the Appendix, due to regularization ambiguities when Weyl fermions are present, the Jacobian in Eq. (4.5) depends on an arbitrary parameter a (see Eqs. (A.15) and (A.19)). The relevance of this parameter was stressed by Jackiw and Rajaraman¹⁹ in their solution of the chiral Schwinger Model; in fact, one can show that if $a > 1$ this model is unitary, Lorentz invariant and consistent. Analogous results can be found for chiral QCD.^{25–26}

For certain nonlinear sigma models arising from the bosonization of fermionic models, one can see that an appropriate choice of a renders the model conformal invariant.^{27–28}

For the σ -models we discuss here, all previous works use an $a = 0$ regularization^{1,9–10} (giving the consistent anomaly). It should be interesting to study the role of the different values of a in the properties of the effective theory, and we hope to come back to this point in a future work. One should notice however, that the proposal we present in this work on how to get an anomaly free model, is valid for arbitrary value of a .

It has been observed^{8–10} that anomalies in sigma models can be cancelled in certain cases by adding ϕ -dependent counterterms so that the variation of the resulting effective theory is zero. This approach can be related with the one we have presented as follows. Consider that one performs in (4.5) the integral over f :

$$\int \mathcal{D}f J^{-1}[f^{-1}, A[\phi]] = \exp(i[\Gamma_{\text{eff}}[A[\phi]] - \Gamma[A[\phi]]) = \exp(i\Delta[A]). \quad (4.10)$$

Then, $\Delta[A]$ in (4.10) plays in our approach, the same role as counterterms constructed in Refs. 8–10. Note however, that in these last works, since counterterms are added by hand, they can be constructed using some representation r for scalars which is, in principle, different from the one selected for fermions. Then, a condition relating a symmetrized trace of generators in both representations has to be satisfied in order to ensure cancellation of the anomaly.

Let us finally comment a third possible approach lying half the way between the one presented here and that developed in Refs. 8–10. In Faddeev-Shatashvili¹² spirit, anomalous theories can be rendered consistent by adding not just a counterterm but new physical fields (the f -fields in our approach) with an adequate Wess-Zumino action. The corresponding generating functional should be obtained starting from the Hamiltonian approach and integrating over momentum variables. One then has to consider that half of the $\dim H$ added variables correspond to coordinate variables and half to momentum variables. Integration over f has to be performed with an adequate Liouville measure Ω^{12} which, in the 2-dimensional example we have dis-

cussed, can be very simply identified with the closed 2-form

$$\Omega = \int dx \operatorname{tr} f^{-1} df \wedge f^{-1} df f^{-1} \partial_x f. \quad (4.11)$$

In terms of Ω , $\omega[f]$ in (3.10) can be written in the form:

$$\omega[f] = \int dt d^{-1} \Omega \quad (4.12)$$

with d^{-1} an “antiderivative” defined as in Ref. 23.

5. Conclusions

We have presented an approach to quantization of nonlinear sigma models defined on G/H spaces which leads to a consistent quantum theory even when internal anomalies are present. We have shown that a careful treatment of subgroup H -degrees of freedom leads to a G -invariant model with H -fields playing a dynamical role. This is our main result, summarized in Eq. (4.5) where the G -invariant effective action Γ_{eff} is explicitly given. The presence of the Fujikawa Jacobian in Γ_{eff} ensures the cancellation of the anomalous variation of the fermionic measure, in the same way the Wess-Zumino action does in chiral gauge theories.¹²

In fact, our approach puts quantization of nonlinear sigma models with other anomalous theories (gauge and gravitational theories with Weyl fermions,^{13–16} chiral boson models²⁰). This point of view for quantization is inspired in Polyakov’s treatment of string theories:¹¹ a degree of freedom, classically decoupled, reappears at the quantum level, with a nontrivial action (the Liouville action in the string case, the Wess-Zumino action given by $\log J[f, A]$, Eq. (3.10), in the sigma model case).

Our approach should be compared with the one developed in Refs. 8–10, where appropriate counterterms are constructed so as to cancel the anomaly. This method basically differs from the one we have presented, since in our case the Wess-Zumino action is not added by hand but naturally emerges in the process of quantization. One can however make contact between both approaches, as we explain at the end of Sec. 4.

The method we have developed is intended to work in arbitrary d space-time dimensions ($d = 2, 4, \dots$) but we have specialized, for simplicity to $d = 2$. It is precisely this case which is relevant for string theories⁴ and in this context, it should be interesting to analyze how anomaly cancellation and conformal invariance are realized. We hope to report on this subject elsewhere.

Acknowledgment

E.M. is partially supported by CONICET. C.v.R and F.A.S. are partially supported by CIC, Buenos Aires, Argentina.

Appendix The Fermion Jacobian

We shall compute in this Appendix, the Jacobian relating fermion determinants

$$\det D_0 = J[f, A] \det D_f \quad (\text{A.1})$$

with

$$D_0 = i\not{D} + i\not{A}P_+ \quad (\text{A.2})$$

$$D_h = h^{-1}D_0h, \quad (\text{A.3})$$

$$P_{\pm} = (1 \pm \gamma_5)/2 \quad (\text{A.4})$$

and

$$f = \exp(i\alpha) = \exp(i\alpha^a H_a). \quad (\text{A.5})$$

(As explained in the text, we have added to the Dirac operator a free left-handed part in order to have a well-defined eigenvalue problem¹⁸). Let us define a one-parameter family of H -transformations interpolating between the identity and $\tilde{h}$, as t ranges from 0 to 1:

$$\tilde{h}_t = e^{i\alpha P_+ t}, \quad h_t^{-1} = e^{-i\alpha P_- t}. \quad (\text{A.6})$$

It is easy to check that

$$\tilde{D}_{h_t} \equiv h_t^{-1}D_0\tilde{h}_t = D_{f_t} \quad (\text{A.7})$$

and hence

$$\det D_0 = J[t] \det \tilde{D}_{h_t}; \quad J[t] \equiv J[f_t, A]. \quad (\text{A.8})$$

Now, since the right-hand side in (A.8) is t -independent, we obtain:

$$J[1] = J[0] \exp\left(-\int_0^1 \omega'(t) dt\right) \quad (\text{A.9})$$

with:

$$\omega'(t) = \frac{d}{dt} \log \det \tilde{D}_{h_t} = \frac{d}{dt} \text{tr} \log \tilde{D}_{h_t} \quad (\text{A.10})$$

or

$$\omega'(t) = i \operatorname{tr} \gamma_5 \alpha. \quad (\text{A.11})$$

Expression (A.11) is an ill-defined quantity and it has to be evaluated using some regularization prescription. Using the heat-kernel method¹⁷ we define:

$$\omega'(t)|_{\text{Reg}} \equiv \lim_{M^2 \rightarrow \infty} i \operatorname{tr} \gamma_5 \alpha \exp(-D_R^2/M^2) \quad (\text{A.12})$$

where

$$D_R = \tilde{D}_{h_t} + a(h_t^{-1} i \partial \tilde{h}_t + i h_t^{-1} A \tilde{h}_t) P_- \quad (\text{A.13})$$

with a an arbitrary parameter introduced so as to take into account the ambiguities that arise when Weyl fermions are present.^{19,24}

Expanding the exponential in (A.11), rescaling variables and taking the $M^2 \rightarrow \infty$ limit one gets:

$$\omega^1(t)|_{\text{Reg}} = -\frac{i}{4\pi} \operatorname{tr} \int d^2x \gamma_5 \alpha D_R^2. \quad (\text{A.14})$$

After some tedious but straightforward calculation one finally obtains:

$$\log J[f, A] = \omega[f] + \frac{1}{4\pi} \operatorname{tr} \int d^2x (A + f \partial_- f^{-1} + a(A_- A_+^f - A_- A_+)) \quad (\text{A.15})$$

with:

$$\begin{aligned} \omega[f] &= \frac{1}{8\pi} \operatorname{tr} \int d^2x (\partial_\mu f^{-1} \partial^\mu f) \\ &+ \frac{1}{12\pi} \operatorname{tr} \int_B d^3x \varepsilon^{ijk} (f^{-1} \partial_i f) (f^{-1} \partial_j f) (f^{-1} \partial_k f) \end{aligned} \quad (\text{A.16})$$

the Wess-Zumino-Witten action.

Note that making $\varepsilon(x) \equiv \alpha(x)$ infinitesimal, one obtains the expression (3.4) for $J[\varepsilon]$:

$$J[\varepsilon] = \frac{1}{4\pi} \operatorname{tr} \int [\partial_- A_+ - a(\partial_+ A_- + [A_+, A_-])] \varepsilon d^2x. \quad (\text{A.17})$$

It is interesting to note that in obtaining (A.15), we have used the transformation (A.6) which corresponds to the evaluation of the change in the fermion determinant when $A_+ \rightarrow A_+ f$ (A_- plays no role in this transformation since the fermions are right-handed). One must note however that in defining a right-handed determinant,

one adds free left-handed fermions. Due to its presence, A_- appears in the final answer (Eq. (A.15)).

Precisely due to the dependence of the right-handed determinant on A_- , one gets a different dependence on a for the Jacobian if one computes J as the ratio:²⁵

$$J[f, A] = \left[\frac{\det \not{D}[A]}{\det i \not{\partial}} \right]^{-1} \cdot \left[\frac{\det \not{D}[Af]}{\det i \not{\partial}} \right].$$

The two alternative results correspond to different regularization choices. (It is well-known that due to regularization ambiguities, the property $\det AB = \det A \cdot \det B$ is not always valid, see for example, Ref. 28). One can pass from one result to the other by an appropriate choice of a dependent counterterms.

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